HW 9

Due on 17 April 2026 at 8PM on Gradescope.

This assignment is worth 25 points, and you should be spending roughly 10 minutes per point on average. If you find yourself spending more than $10n$ minutes on a n-point problem, reach out to me via drop-in hours, Discord, or email.

Collapse all exercises

Goals

- Finish your final homework assignment! :)
- Review previous assignments
- Work on your blog post
- Reflection and exit interview
- Practice with group actions and rings

▼ EXERCISE 1 [2]

1. Fill out the following and tag it as Exercise 1:

```
\section*{Exercise 1}
Collaborators: Alex, Ross, Margaret, Joan

\noindent
Tools and resources used: On Exercise 1, I used ChatGPT to help
\pagebreak
```

You don't need to include class notes or textbooks under "tools and resources."

2. On *each* individual exercise you attempt (both required and choice) start with a header based on this template. (Make sure that the denominator of the "self-assessed score" matches the number of points attempted.)

```
\section*{Choice Exercise 5}
\begin{itemize}
  \item Points attempted: 7
  \item Self-assessed score: 6/7.
  \item Time spent: 56 minutes.
  \item Difficulty: 8/10.
  \item Questions/Feedback: None.
\end{itemize}
```

3. Choose one or more problems that you would most like feedback on, and indicate this under the "questions/feedback" bullet with a brief note of what you would most like me to focus on. (This could be "mathematical correctness," "grammar," "notation," "formatting," or anything else.)

▼ EXERCISE 2 [2]

Go through the previous homework assignments (including this one) spending at least a minute browsing through each one. Make a note of one or two problems that stood out for you this term, which you would recommend to future students taking this class, and write a note about why. These might be problems where something "clicked" for you, where it helped you to make a connection to new ideas, where you generated a diagram that was meaningful, etc.

Post on the `#hw-prompts` Discord channel about which problems you chose and why.

▼ EXERCISE 3 [6] (BLOG POST)

Set a timer for 60 minutes and work on your blog post. Your goal is to get the blog post in a Markdown format which (after further edits) will be ready to be posted on the class website. To do this, you're going to do the following:

1. Log into Github. (You will need to create a Github account if you haven't done this before.)
2. Navigate to https://gist.github.com/. Where it says `Filename including extension...`, you will type `blog_post_draft.md`. In the body below, you will post your existing markdown that you have been working on.
3. Make sure your blog post has an image or another piece of media included, you can upload this with the `Add file` button at the bottom.
4. Hit "preview" to make sure it looks how you want it to.
5. When ready hit "Create secret gist." Send me the URL from the new page, so that I can view it. (This draft is editable, so you can keep iterating on it.)
6. Spend any remaining time working on your draft.

Choice Exercises

Do 15 points of the following.

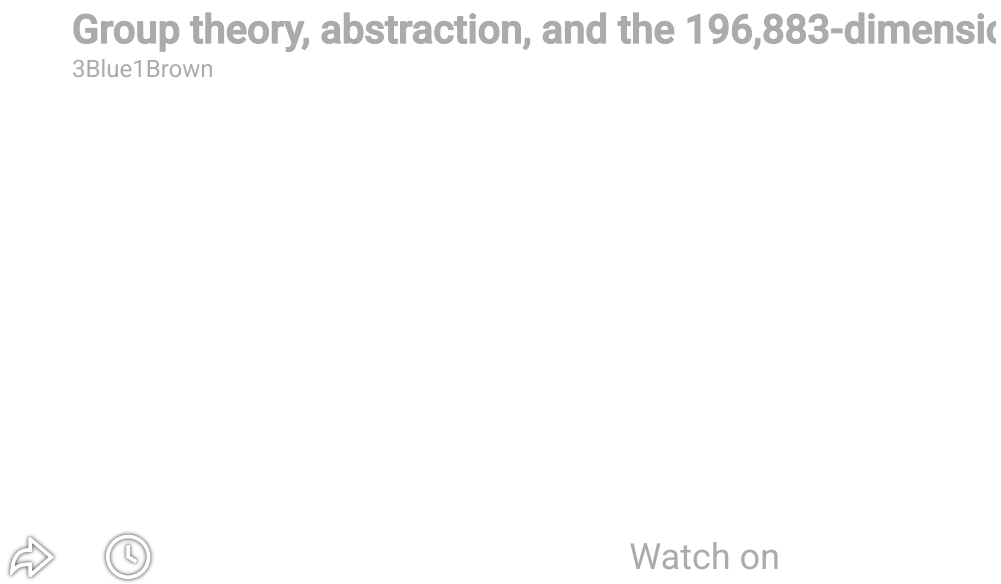
▼ CHOICE EXERCISE 4 [4] (GOING DEEPER)

Set a timer for 20 minutes. Click on the links to read some of the highest voted posts on Math Stack Exchange with the `[abstract-algebra]` or `[group-theory]` tags. Don't be afraid to dive into posts containing words that you're not familiar with, but also don't hesitate to move on to a different post if you're feeling lost. When the timer goes off, make a note about something you learned or something you are curious about.

▼ Choice Exercise 5 [4] (reflection)

At the beginning of the term, I had you watch the 22-minute 3Blue1Brown video Group theory, abstraction, and the 196,883-dimensional monster. In this exercise you're going to watch it again.

1. Before you watch the video again, go look at your HW 0 Exercise 3 response and re-read it, and write down at least one though.
2. Watch the video!



3. While you watch the video, make a list of ideas from the video that you understand better after having taken this class *and* list ideas that you still don't fully understand.
4. Write a sentence or two reflecting on your learning experiences this term.

▼ Choice Exercise 6 [≤5] (Exit interview)

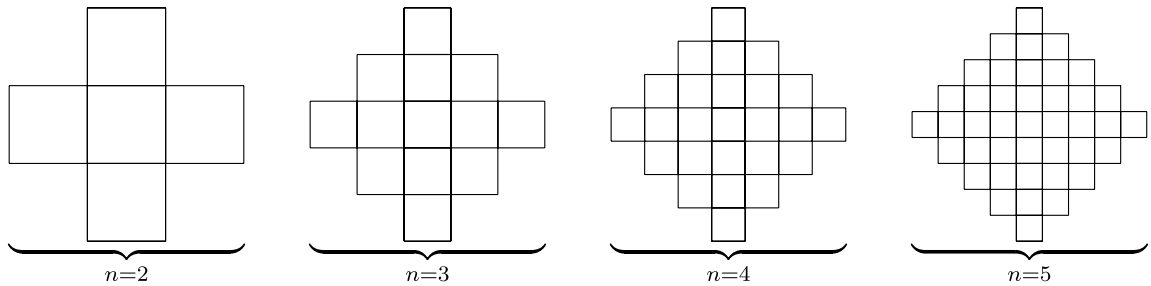
It's possible that you might ask me for a letter of recommendation in the future, but it might be in many months after you've forgotten a lot of details of your experience in this class. Write a bulleted list of work that you're proud of this term, ways you think you've been a good classmate or student, examples of ways in which you've flourished or grown, etc. (You get one point for every two

items in your list, up to a maximum of five points.) Aim to spend about 3–5 minutes on each bullet point.

(If you ask me to write you a letter of recommendation in the future, please reference this list.)

▼ Choice Exercise 7 [8] (Burnside's lemma)

This problem will have you solve a natural problem in combinatorics where you will count something that I don't *think* anyone has ever counted before: how many ways are there to fill the *diagram for the n-th centered square number* (illustrated below) with tiles that consist of either a horizontal or vertical stripe: and



Illustrations of the diagrams of size n.

The diagram for the n-th centered square number can be defined by placing unit squares centered at all points $(x, y) \in \mathbb{Z}^2$ satisfying $|x| + |y| < n$. Let $a(n)$ be the number of ways of tiling the diagram for the n-th centered square number up to the symmetries of D_4, which are the rotations and reflections of the square.

1. Prove by induction or any other method that the number of squares in the diagram is $n^2 + (n-1)^2$, and describe the diagram for the first centered square number.
2. Compute $a(1)$ by "brute-force" by drawing all of the possible ways of tiling the square up to rotation/reflection.
3. Compute $a(2)$ using Burnside's lemma. (Note, you might just do the next part first.)
4. Give a formula for $a(n)$ using Burnside's lemma, and check that it specializes to give the correct value for $a(1)$.
5. Compute explicit numerical values for $a(3)$, $a(4)$, and $a(5)$. (Using a calculator like WolframAlpha might be helpful here!)

6. $a(n) \neq 2^{n^2+(n-1)^2-3}$, but it's an "asymptotically close" approximation. What is the relative error between $a(5)$ and the estimation? Which is bigger, $a(5)$ or $2^{n^2+(n-1)^2-3}$?

7. Look up the sequence $a(1), a(2), a(3), a(4), a(5)$ on the On-Line Encyclopedia of Integer sequences, and see if it's there!

▼ CHOICE EXERCISE 8 [≤10] (FAITHFUL GROUP ACTIONS)

We call a group action **faithful** if for all nonidentity elements $g \in G \setminus \{e\}$, there exists $x \in X$ such that $g \cdot x \neq x$. Informally, every non-identity element *does something*.

1. [3] Let $X = \{1, 2, 3, 4, 5\}$ and let $G = C_6$. Give an example of a faithful group action of G onto X. (Hint: is there a one-to-one homomorphism ϕ: $C_6 \to S_5$?)

2. [2] Let $(G, *)$ be a group that acts on a set X and let

$$N = \{g \in G \mid g \cdot x = x \text{ for all } x \in X\}.$$

In class, we showed that N is a normal subgroup. Repeat the argument here.

3. [3] We care about normal subgroups because we can *quotient by them*. Show that G/N acts on X (denoted $\bullet$) by

$$gN \bullet x = g \cdot x,$$

and that this map is well-defined: if $g_1 N = g_2 N$ then $g_1 \cdot x = g_2 \cdot x$.

4. [2] Prove that the action of G/N is faithful.

▼ CHOICE EXERCISE 9 [6] (GROUP RING)

In class we let $G = D_3$ be the dihedral group of the triangle, and did a warmup with the group ring $\mathbb{C}[G]$, where each element is written as a "formal sum" $\sum_{g \in G} z_g g$ where $z_g \in \mathbb{C}$ is a complex number.

Recall that we add these elements the same way we add polynomials, where we combine like terms. When we multiply these elements, it's like multiplying polynomials, but where we "multiply" the elements of G using the binary operation (and order matters!) In this problem let $x, y, z \in \mathbb{C}[G]$ be given by

$$x = 3r - 2f, \quad y = -1e + 1r^2 f, \quad z = 1e + 2f$$

1. Compute $x + y$ and $x + y + z$.

2. Compute $z \times x$ and $z \times y$.

3. Compute $z \times (x + y)$

4. Compute $z \times x + z \times y$.

5. Since $\mathbb{C}[G]$ is a ring, $(\mathbb{C}[G], +)$ is an abelian group. What is the additive identity 0_R in R? What is the inverse of the element $\sum_{g \in G} z_g g$?

6. Since $\mathbb{C}[G]$ is a ring, $(\mathbb{C}[G], \times)$ has a multiplicative identity. What is the multiplicative identity 1_R in R? Prove that is the multiplicative identity by computing $1 \times (\sum_{g \in G} z_g g)$.

7. Is there an element $w \in \mathbb{C}[G]$ such that $w \times 3r = 1_R \in R$?

8. Is there an element $\alpha \in \mathbb{C}[G]$ such that $\alpha \times x = 1_R \in R$? (Set a timer for 5 minutes and try to find one. If you find one, what is it? If you don't find one, do you think that x has a multiplicative inverse?)

▼ Choice Exercise 10 [4 + 4] (Rings?)

1. [4] Given two matrices $A, B \in \mathbb{R}^{n \times n}$ with entries $\{a_{ij}\}$ and $\{b_{ij}\}$ respectively, we define matrix multiplication as $AB = C \in \mathbb{R}^{n \times n}$ such that $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$. Use this definition to prove that matrix multiplication distributes over matrix addition.

2. [4] Let $(G, \oplus)$ be an abelian group, and denote

$$\mathrm{Endo}(G) = \{\phi \colon G \to G \mid \phi \text{ is a homomorphism}\}.$$

Prove or disprove that $\mathrm{Endo}(G)$ is a ring where the addition is pointwise and multiplication is function composition.

$$(\phi_1 + \phi_2)(g) = \phi_1(g) \oplus \phi_2(g)$$
$$(\phi_1 \times \phi_2)(g) = \phi_1(\phi_2(g))$$

▼ Final Choice Exercise [≤7]

Do any subset of the following parts, state which you did and how many points you are claiming for this problem (maximum of 7 points).

1. [≤3] Email Prof Kagey (pkagey@cpp.edu) or post in the `#feedback` channel on Discord with a typo that you have spotted in this assignment or with a

suggestion about how to make something clearer. (One point for every six minutes that the typo got you stuck, with a maximum of three points.)

2. [≤2] Ask a question on Discord. (One point per question, up to 2.)

3. [≤4] Answer a classmate's question on Discord or give a hint about a problem on this week's assignment.[*] (Two points per answer/hint, up to 4.)

4. [≤4] Can you generalize any of the above problems in an interesting way? If so say what the generalization is, attempt to solve it, say something about it, and say how many points you feel like your effort was worth, up to 4.

5. [≤4] Choose one or more problems from our textbook or a previous homework assignment which interest you and write the problem statement along with an answer. Say how many points you feel like your effort was worth, up to 4.

6. [2] Stop by student drop-in hours.

7. [2 or 5] Work on at least one of the above exercises with a classmate. State which classmate you worked with and which exercises you collaborated on. If you've collaborated with someone for the first time, you get 5 points, if you've collaborated with someone before, you get 2 points for working with them again.

8. [2] Turn in the assignment on time (without requesting an extension).

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[1] The syntax for a hint on Discord is double bars `||` on each side of the spoiler like this: `Remember that ||1 + x < x + 2||.` ↩

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